B.E. INFORMATION TECHNOLOGY • FINAL YEAR PROJECT • 2026–27

Clinexa

A Unified AI-Powered Clinical Decision Support
& Healthcare Management Platform

Complete Project Reference Guide (Easy Language)

Prepared for: Sachin Latke, Mahesh Suryawanshi, Asif Shaikh
Guide: Yogita Rode
Department of Information Technology
NMVPM's Nutan Maharashtra Institute of Engg. & Tech., Talegaon

1. What is Clinexa?

Clinexa is one app that connects **patients, doctors, and hospital staff** on a single platform, so all three sides can see the same information and get AI help — instead of using separate, disconnected tools.

A patient can describe symptoms to an AI chatbot in their own language, get a possible diagnosis with a clear explanation, find nearby hospitals for that condition, and book an appointment — all in one place. The doctor sees the same case with full context before the visit. The hospital gets a live dashboard of beds, resources, and disease trends.

> **In one line:** Clinexa is an AI health assistant that helps patients get quick, understandable guidance, helps doctors decide faster with clear AI reasoning, and helps hospitals manage resources — all connected together.

2. The Problem We Are Solving

1. Patient records are scattered

A patient's health history is spread across many clinics and hospitals. No single place has the full picture.

2. Doctors are overloaded

Doctors see too many patients and have very little time to check symptoms, history, and medicines carefully.

3. People trust the wrong sources

Before seeing a doctor, patients often Google their symptoms or ask on social media, which causes fear or wrong ideas.

4. Hospitals can't see the full picture

Hospital staff have no live dashboard to know how many beds are free or which diseases are increasing.

3. Our Goals (Objectives)

1. Build an AI tool that checks a patient's symptoms and gives simple, easy-to-understand health advice.
2. Help doctors by suggesting possible illnesses and warning them about medicine problems, so they can decide faster.
3. Keep one safe, complete health record for each patient that can be seen anywhere they go for treatment.
4. Give hospital staff a live dashboard showing beds, resources, and disease trends, all in one place.
5. Always explain why the AI gave a certain suggestion, so doctors and patients can trust it.
6. Let people use the app in their own regional language, so it is easy for everyone in India to use.

4. What Others Have Tried (Literature Review)

Study	What it did	What we learned
eSanjeevani AI Health Chat (AIIMS Rishikesh & Wadhwani AI, 2025)	Used AI to understand symptoms in 12 Indian languages for online doctor visits	Works well for basic sorting of patients, but does not give a full diagnosis
Study on Medicine Support Tools (Lehmann & Subbiani, 2024)	Looked at tools that warn doctors about medicine problems	Good results, but tested in only one hospital at a time
Do Doctors Trust AI Tools? (Tun et al., 2025)	Reviewed 27 studies on how much healthcare workers trust AI	Doctors trust AI more when it explains its reasoning and fits their daily work
AI Tools in Healthcare, A Review (Elhaddad et al., 2024)	Reviewed many AI tools used in hospitals	Connecting new AI tools to old hospital systems is still hard
Explaining AI Decisions in Healthcare (2024)	Studied 62 papers on making AI explain itself	Explaining AI builds trust, but is still difficult to do well

The gap Clinexa fills

Most existing tools solve only one problem at a time — some help patients, some help doctors, some help hospitals, but not all together. Most are tested in a single hospital and don't work well elsewhere. Few explain their answers clearly or work in local languages. Clinexa brings patients, doctors, and hospital staff onto one platform, where every AI answer is explained clearly and works in many languages.

5. How Clinexa Works — Full Patient Journey

This is the complete, updated flow from the moment a patient opens the app to the moment they see a doctor, including the appointment booking and hospital recommendation features.

1. Patient describes symptoms to the chatbot, by typing or talking, in their own regional language.
2. The AI (language understanding model) reads the message and picks out the important health details.
3. The AI (prediction model) lists the most likely conditions and explains why, with a confidence level.
4. **New:** Based on the suggested disease, the app finds nearby hospitals that have the right department/specialist available.
5. **New:** Patient sees a list of nearby matching hospitals and can pick one to book an appointment directly, or book with a specific doctor if they already know who they want to see.
6. The doctor logs in separately and sees the appointment along with the patient's AI-generated case summary and full health history — before the visit.
7. The doctor confirms or edits the diagnosis and writes a prescription.
8. The record is saved to the patient's permanent health file, and the hospital dashboard updates automatically to show overall trends and appointment load.

Why Clinexa is different

- It helps patients, doctors, and hospitals all at once, instead of just one group.
- It always explains its answers clearly, which helps people trust it more.
- It can connect with hospital systems that already exist, instead of replacing them.
- It doesn't just diagnose — it guides the patient to an actual next step (a nearby hospital and a booked appointment).

6. Appointment Booking Feature Added

Patients are not limited to only chatting with the AI — they can also book an appointment directly.

Two ways a patient can book

- **Through the chatbot:** after getting a diagnosis suggestion, the app recommends nearby hospitals with the right specialist and lets the patient book right there.
- **Directly:** if a patient already knows what they need (say, a follow-up visit), they can skip the chatbot and book a slot straight away.

What happens behind the scenes

- **Disease-to-department mapping:** a simple lookup table that connects a disease to the right specialist (e.g., skin issue → Dermatology, fever/dengue → General Medicine). This is plain logic, not a machine learning model.
- **Hospital database:** stores each hospital's name, address, location, departments offered, and bed/slot availability.
- **Location service:** finds hospitals near the patient using their current location (can use a mapping service like Google Maps Platform, or simple distance calculation if the hospital list is small).
- **Filtering and sorting:** shows hospitals that have the needed department, sorted by distance, and optionally by current bed availability.
- **Booking + slot check:** checks the doctor's real-time availability, prevents double-booking, and confirms the slot.

> **Optional smart touch:** instead of just showing every open slot, the system can suggest the earliest reasonable slot based on urgency — if the AI's confidence in a serious condition is high, it prioritizes sooner appointments. This can be called "AI-assisted scheduling" without needing a heavy ML model — a simple priority rule is enough.

7. Doctor's Side — What They See Added

Doctor's flow

1. Doctor logs in to their own separate doctor dashboard.
2. Doctor sees a list of patients waiting for review — from chatbot conversations or booked appointments.
3. Doctor opens a specific patient's case to see full details.
4. Doctor reviews everything and adds their own diagnosis/prescription.
5. This is saved back to the patient's record, visible to the patient as well.

What exactly is inside a patient's case view

- **Patient's symptoms** — the original conversation with the chatbot, translated if needed.
- **AI's suggested diagnosis and confidence level** — e.g., "Dengue (70%), Viral fever (20%)."
- **AI's explanation** — which specific symptoms led the AI to that conclusion.
- **Patient's medical history** — past visits, diagnoses, medications, allergies, pulled from their unified health record.
- **Appointment details** — time slot and reason for visit, if booked.

What needs to be built for this

- **Doctor authentication** — a separate login role so the system shows the doctor dashboard, not the patient app.
- **Role-based access control** — the system checks "is this a doctor, patient, or hospital admin?" and only shows the relevant screens and data to each. This is an important point to mention in your project.
- **A single case view** — one screen pulling together symptoms, AI diagnosis, explanation, history, and booking info instead of scattering it across screens.
- **Queue management** — sorting which patients the doctor should see first, by appointment time or by urgency if the AI flags something serious.

8. System Architecture

Clinexa is built in layers, where each layer has one clear job, and data flows from top to bottom:

1. What Users See

Patient App • Doctor Screen • Hospital Admin Screen

2. The Connecting Layer

Handles logins for all three user types (role-based access) • Sends data safely between all the parts • Manages appointment booking requests and hospital searches

3. The AI Brain

Reads symptoms • Predicts illness with a confidence score • Explains its answer • Understands many languages • Suggests the right hospital department

4. Where Data is Stored

Patient health records • Chat history • Hospital statistics • Hospital directory (location, departments, availability) • Appointment/schedule data

5. Outside Connections

Labs • Pharmacies • Government health systems • SMS alerts • Maps/location service

9. The AI Models — Explained Simply

Clinexa does not use one single AI model. It needs several models working together, each doing a different job:

1. Language understanding model (NLP model)

Reads what the patient types or says, in their regional language, and pulls out the useful medical information — which symptoms, since when, how severe. It turns messy human sentences into structured medical clues.

2. Disease/illness prediction model

Takes the structured symptoms and predicts the most likely conditions, with a confidence score for each. Built using **Python, TensorFlow, and scikit-learn** — TensorFlow for deep learning (neural network) approaches, scikit-learn for simpler, faster classical ML models (like decision trees or random forests). Both can be tried, and whichever performs better on the project's dataset is kept.

3. Explainability model (the "why" model)

This is what makes Clinexa different from most other tools. Instead of just saying "it might be dengue," it also shows why — which symptoms mattered most. Usually done with a technique like **SHAP** or **LIME**, which open up a "black box" ML model and show which input features drove the decision. This builds doctor trust, matching what the literature review found.

4. Medicine-interaction checker

A smaller supporting model or rule engine that flags if a suggested medicine could clash with something the patient already takes, warning doctors before they prescribe.

5. Disease-to-hospital-department mapping

Plain logic (not a machine learning model) that connects a predicted disease to the right specialist/department, used for the hospital recommendation feature.

6. Multi-language support

Not exactly a separate "model," but a translation/localization layer so patients can type or speak in their own regional Indian language and the AI still understands.

10. Technology Stack

Layer	Tools Used
Screens (Frontend)	React.js for the website • Flutter for the mobile app
Backend (Server)	Node.js or Django to run the app's logic
AI & Smart Features	Python with TensorFlow and scikit-learn • Explainability tools (SHAP/LIME)
Storing Information	One database for health records • another for chat messages • hospital and appointment data
Connecting to Hospitals	Standard health data format so it works with existing hospital systems
Location / Maps	A mapping service (e.g., Google Maps Platform) for finding nearby hospitals
Hosting Online	Cloud hosting (AWS/Azure) • Tools to deploy updates smoothly
Keeping Data Safe	Secure logins • Encrypted data • Follows India's health data rules

11. Implementation Plan — 16 Weeks

Phase	Weeks	Focus	What Happens
Phase 1	1–3	Planning	Study past research, gather requirements, find good data to use
Phase 2	4–6	Building the AI	Clean the data and build the AI model that predicts illness
Phase 3	7–9	Building the Backend	Build the server, database, and health record system, including hospital and appointment data
Phase 4	10–12	Building the Screens	Create the app screens for patients, doctors, and admins, including booking and hospital search
Phase 5	13–14	Connecting Everything	Add AI explanations, multiple languages, hospital connections, and role-based access
Phase 6	15–16	Testing & Launch	Test everything carefully, fix issues, and launch the app

12. Expected Outcome & Deliverables

- A working app with separate, easy screens for patients, doctors, and hospital staff.
- An AI helper that suggests possible illnesses and clearly explains its reasoning.
- Faster first checkup for patients, since AI helps sort symptoms quickly.
- A live dashboard that helps hospitals plan beds, staff, and resources better.
- An app that works in regional languages, so more people across India can use it.
- A hospital recommendation and appointment booking system that gives patients a clear next step after diagnosis.
- A strong base that can grow and connect with India's national digital health system.